Mithil Rajapakse

Electrical & Electronic Engineering Undergraduate

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PROFESSIONAL SUMMARY

Final-year Electrical & Electronics Engineering undergraduate with hands-on experience developing AI, automation, and software solutions for real-world engineering and business problems. Experienced in Python, RAG/LLM systems, machine learning, backend APIs, and workflow automation, with projects spanning explainable AI, graph-based retrieval, and engineering process automation. Strong communicator with proven leadership and experience translating technical concepts into practical solutions.

WORK EXPERIENCE

Hayleys Fentons Limited – Solar Division | Nov 2025 – Feb 2026

Electrical Engineering Intern (Solar PV Projects)

Worked on utility-scale and industrial rooftop solar PV projects ranging from 300 kW to 1.5 MW, including sites at Alumex Ekala, Captain Steel, Waguruwela Oil Mills, and Femstar Packaging.

- Independently identified a manual, error-prone Bill-of-Materials process for solar installations and built a Python tool (Tkinter GUI + CLI) to automate it, parsing rooftop layouts to auto-generate material lists, cable sizing, and inverter counts, with Excel report export.
- On own initiative, developed a combinatorial optimization tool to automate solar panel string-configuration design, replacing manual spreadsheet trial-and-error with an algorithm that ranks layouts by installation practicality and inverter load balance.

TECHNICAL SKILLS

AI/Data/Automation

- RAG
- Graph-based retrieval
- Vector search / embeddings
- LLM integration
- OpenAI / Gemini / OpenRouter workflows
- Explainable AI
- Python automation
- pandas
- numpy
- scikit-learn
- data analysis
- workflow automation
- n8n
- MCP integrations

Electrical/Embedded

- C
- Embedded C
- Arduino
- TI MSP430
- Assembly
- MATLAB
- Quartus II
- Multisim
- LTSpice
- AutoCAD
- circuit design
- microcontroller programming
- control systems

Technical Skills

- Python
- FastAPI
- SQLAlchemy
- PostgreSQL
- REST APIs
- HTML/CSS/JavaScript
- Jinja2
- Docker
- Git/GitHub
- Linux / shell
- pytest

Business/Product

- WordPress
- WooCommerce
- PHP
- custom plugin development
- web product workflows
- process automation
- technical documentation

LEADERSHIP

Head of PR - IEEE Student Branch of Curtin University Colombo (2025 Nov - Present)

Founding committee member; leading a PR team of 16+, owning publicity end-to-end for all chapter events and managing the branch's presence across all socials.

President - Curtin Colombo Student Guild 24/25 (2024 Nov - 2025 Nov)

Elected head of the university's official student representative body, leading a team of 150+ students; created the Houses system and organized the university's first-ever Sports Meet (Eclipse '25).

Asst. Treasurer - Curtin Colombo Student Guild (2023 Nov - 2024 Nov)

Oversaw finances across the Guild's major campus events; as Head of Finance for Unplugged 2024, the university's largest event (a concert), secured over LKR 1 million in sponsorships.

EDUCATION

Curtin University Colombo | 2023-Present

BEng (Hons) Electrical & Electronics Engineering

Royal College Col.07 | 2011-2023

Combined Mathematics

- GCE A/L's - 1 A 3 C (2022/23) - [General English A]
- GCE O/L's - 9 A (2019)

PROJECTS

Solar PV XAI research project

- Developing an explainable AI framework for solar PV fault diagnosis
- Built a Python-based data pipeline for fault classification and interpretation
- Focused on transparent diagnostics for operations and maintenance decisions
- Applied machine learning + explainability techniques to engineering domain data
- Technologies: Python, pandas, numpy, scikit-learn, ML evaluation, explainability techniques

Graph-first RAG system - by IFS - SLIIT CODEFEST 2026

- Built a retrieval-augmented generation system using a graph-first retrieval strategy with vector fallback
- Designed for evidence-grounded answers across a document corpus
- Emphasized explainability, retrieval traceability, and answer provenance
- Technologies: Python, retrieval pipelines, embeddings, graph retrieval, LLM orchestration

CheckPoint - Full NFC/QR attendance system for university use

- Developed backend APIs, authentication flow, attendance logic, and role-based access
- Designed data model for sessions, attendance, audits, and room tracking
- Technologies: FastAPI, SQLAlchemy, PostgreSQL, Docker, Jinja2, server-side auth

Solar engineering automation tool - Hayleys Fentons | Solar Division

- Built a Python desktop CLI + Tkinter application to automate rooftop solar BOM generation
- Parsed rooftop layouts, calculated cable sizes, inverter counts, and exported Excel reports
- Reduced manual engineering effort and improved consistency in solar project planning
- Technologies: Python, Excel export, GUI/CLI workflow, engineering calculations

Solar string optimizer - Hayleys Fentons | Solar Division

- Developed a combinatorial optimization tool to rank solar panel string layouts
- Improved practicality of installation layouts and balancing across inverter loads
- Replaced manual spreadsheet trial-and-error with algorithmic evaluation

Event Logistics Platform - CV Boost 2026

- Built a backend and frontend application for event logistics and coordination.
- Implemented structured workflows, user roles, and operational process modeling.

UNIVERSITY PROJECTS

2026

- Line Follower Robot: designing and building an autonomous Arduino-based line-following robot, integrating sensor input, motor control, and embedded logic

2025

- Terminal-Based Snake Game – Implemented Snake game in C within Ubuntu terminal.
- Eco-Play Tiles – Proposed low-carbon, oxygen-generating tiles as a cost-effective sustainable innovation.
- 8-bit CPU Design – Built and tested CPU using Quartus II logic gates.
- Op-Amp Simulation – Modeled op-amp circuits with Multisim.
- Microcontroller LED Control – Programmed PCB-based microcontrollers with C and Assembly.

2024

- Garden Lifecycle Simulator – Developed a lifecycle simulation using Python in Linux
- Circuit Design & Simulation – Designed and tested Arduino-based circuits with LTSpice
- Reaction Time Data Analysis – Conducted statistical analysis using Excel on human reaction times.
- Sustainable Garden for Saibai Island – Designed and pitched an innovative 3D design for a garden based food sustainability solution.